

# Field Intelligence Report

Parcel 54000 · Generated 27 July 2026 · LPIS 2024



Selected parcel among surrounding mapped fields (illustrative)

## Crop Detection

| SATELLITE ESTIMATE                                    | LPIS DECLARATION (FARMER)                                                    |
|-------------------------------------------------------|------------------------------------------------------------------------------|
| <div>■ Maize</div> <div>Confidence: 100% ■ High</div> | <div>■ Barley - Spring</div> <div>Status: ■ Satellite estimate differs</div> |

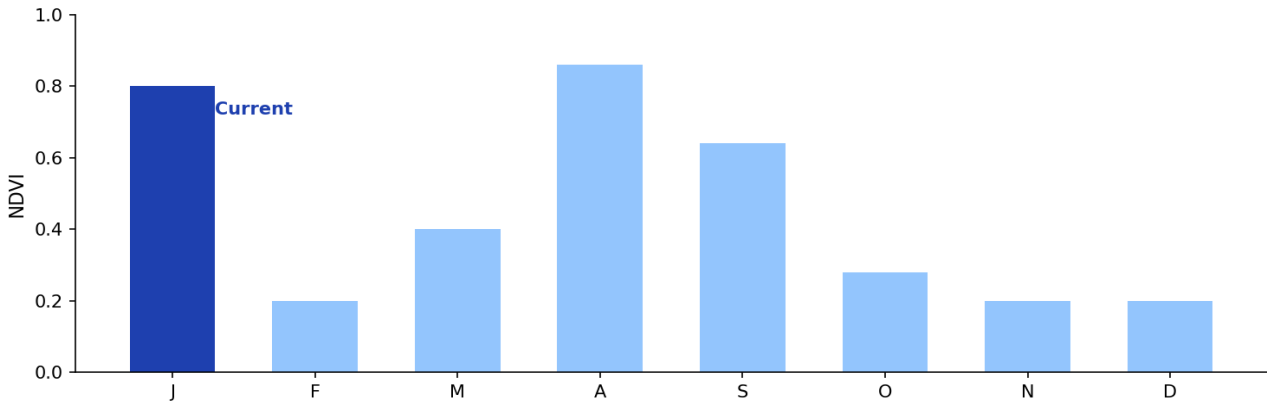
This difference is presented as evidence for review, not as a determination that the declaration is incorrect — see Methodology for how differences are calculated.

|            |                    |                   |                |
|------------|--------------------|-------------------|----------------|
| 1.6 ha     | Silking            | 0.80              | 5–15 Oct       |
| FIELD AREA | GROWTH STAGE (4/7) | NDVI (VEGETATION) | HARVEST WINDOW |

## Growth Timeline

|                |                 |                 |              |            |       |          |
|----------------|-----------------|-----------------|--------------|------------|-------|----------|
| ✓<br>Emergence | ✓<br>Vegetative | ✓<br>Tasselling | ■<br>Silking | Grain Fill | Dough | Maturity |
|----------------|-----------------|-----------------|--------------|------------|-------|----------|

## NDVI Seasonal Profile



Satellite Observations

| Metric                   | Value        | Note                            |
|--------------------------|--------------|---------------------------------|
| NDVI (Vegetation)        | 0.80         | Dense canopy                    |
| Soil Moisture (estimate) | Medium (29%) | Live, per-location — Open-Meteo |
| Cloud cover (7-day)      | ~45%         | Affects imagery freshness       |

7-Day Weather Forecast

|                       |                             |                 |
|-----------------------|-----------------------------|-----------------|
| 6 mm<br>EXPECTED RAIN | 9–22°C<br>TEMPERATURE RANGE | 20 km/h<br>WIND |
|-----------------------|-----------------------------|-----------------|

AI Insights

■ High confidence difference from LPIS

Satellite strongly suggests Maize but the LPIS declaration is Barley - Spring. This parcel may warrant ground verification.

■ Healthy crop

Vegetation indices indicate good canopy development. No signs of stress detected from satellite data.

■ Next satellite pass

HLS revisit in approximately 4 days. Updated NDVI and vegetation maps will be available.

Parcel Information

|                   |                   |                    |                       |
|-------------------|-------------------|--------------------|-----------------------|
| Area              | 1.6 ha (~4 acres) | Parcel ID          | 54000                 |
| Centroid Latitude | 53.1602°N         | Centroid Longitude | -7.0502°W             |
| Source            | LPIS 2024         | Data vintage       | 2022–2025 (4 seasons) |

# Methodology & Disclaimer

## How this report was produced

This crop prediction is generated using a CatBoost machine learning model trained on 61,566 Irish tillage parcels from the LPIS 2024 dataset, using HLS (Harmonized Landsat Sentinel-2) satellite imagery — NDVI, EVI, NDWI, NDRE and NDII vegetation indices — across four growing seasons (2022–2025). The model achieves 77.05% balanced accuracy nationally on held-out validation data. Confidence scores are calibrated using isotonic regression so that the displayed percentage reflects genuine prediction reliability. Soil moisture is fetched live per-location from Open-Meteo at the time each parcel is viewed.

## Reading confidence scores

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| ■ High (80–100%)  | Prediction is usually reliable.                          |
| ■ Medium (50–79%) | Similar crops may also fit the satellite signature.      |
| ■ Low (<50%)      | Use as a prompt for further review, not as a conclusion. |

## Known limitations

- Spring cereals (Barley, Oats, Wheat) are more easily confused with each other than with Maize or winter crops.
- Satellite observations are affected by cloud cover; confidence may decrease when limited clear imagery is available.
- Growth stage and harvest window are model estimates, not field-verified dates.
- The current model covers 8 tillage crop types only; grassland and other land uses are not classified.
- Soil moisture is a live estimate from weather-model data, not a direct field sensor reading.

**Disclaimer:** Cube Earth provides decision-support information and should not be used as the sole basis for regulatory or contractual decisions without additional verification. A difference between satellite prediction and LPIS declaration does not, by itself, establish that either is incorrect.